

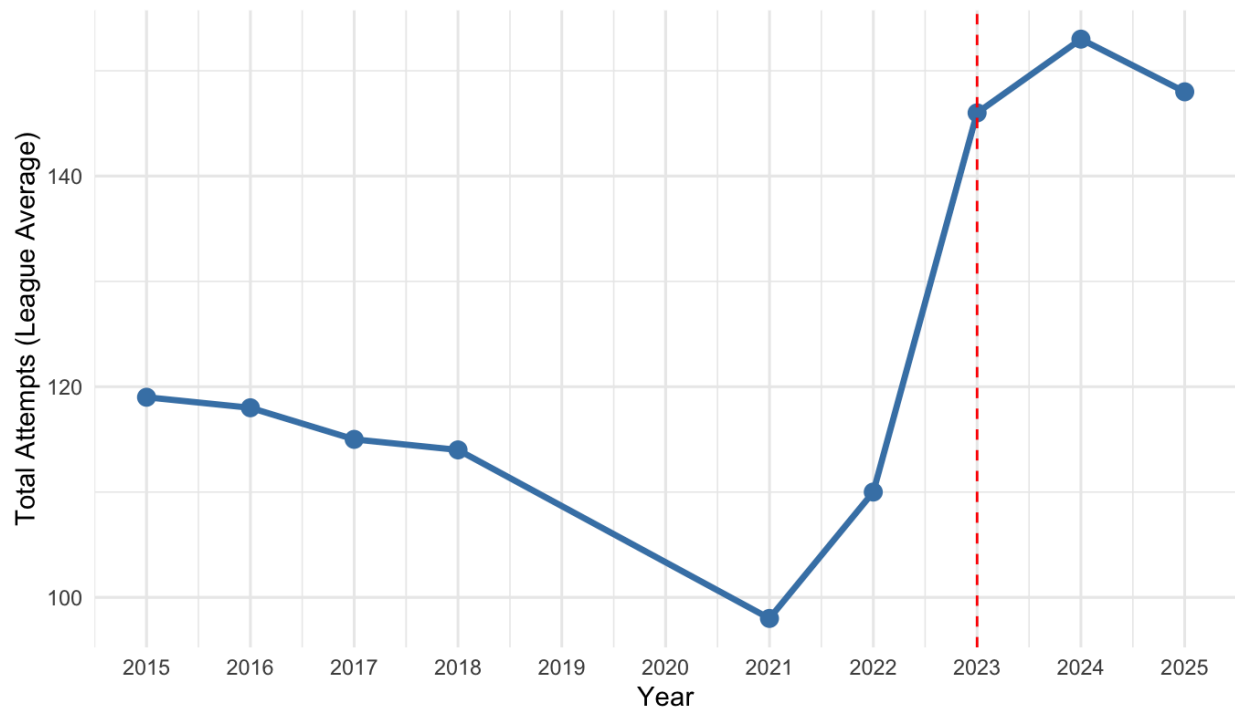
Write up

The MLB implemented a set of rule changes before the 2023 season. Included in these changes were the pitch clock, bigger bases, and shift bans. We hypothesized that these changes, particularly the bigger bases and the banning of the shift, would lead to an increase in baserunning aggressiveness in steal and potential extra-base situations. We gathered data on stolen base attempts, extra-base attempts, and run expectancy pre and post rule change. As we'll discuss, the new rules had a measurable impact on scoring overall.

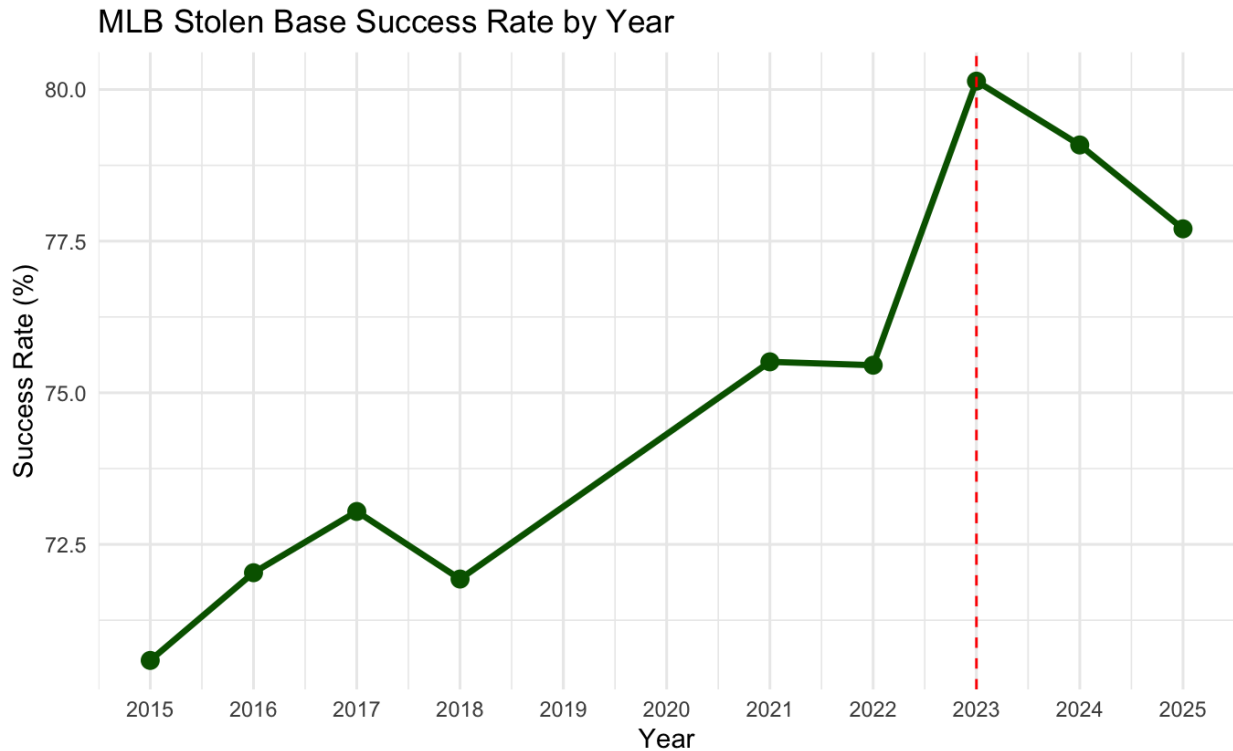
Visualization of Trends

	Year	SB	CS	SB_attempts	SB_pct	rule_change
1	2015	84	35	119	70.58824	Pre
2	2016	85	33	118	72.03390	Pre
3	2017	84	31	115	73.04348	Pre
4	2018	82	32	114	71.92982	Pre
5	2021	74	24	98	75.51020	Pre
6	2022	83	27	110	75.45455	Pre
7	2023	117	29	146	80.13699	Post
8	2024	121	32	153	79.08497	Post
9	2025	115	33	148	77.70270	Post

MLB Stolen Base Attempts by Year



Following the rule changes implemented in 2023, we are able to see a clear jump in the number of stolen base attempts, with average attempts across the league increasing by ~20 attempts per team. The fact that this number not only increased in 2023 (year of the rule changes), but also remained at this higher number indicates that it likely is related to the rule changes, as the number of attempts had never reached a number this high prior to the rule changes.



The same trend is visible when analyzing success rate by year, as we see an instant increase in success rate, reaching a new high of over 80% in 2023, with a small decrease in rate in the years following. While the success rate continues to decrease since 2023, it is still at a higher rate than in years prior. One potential explanation for this is the increase in the size of the bases, as the distance from first to second is now 4.5 inches shorter.

Regression Analysis

Call:
`lm(formula = SB_attempts ~ Year + rule_change, data = yearly)`

Residuals:

Min	1Q	Median	3Q	Max
-8.4816	0.1265	1.0653	1.3224	5.5837

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4329.1796	1579.2584	2.741	0.033679 *
Year	-2.0653	0.7803	-2.647	0.038185 *
rule_changePre	-48.7143	5.7573	-8.461	0.000149 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.986 on 6 degrees of freedom
Multiple R-squared: 0.9505, Adjusted R-squared: 0.934
F-statistic: 57.58 on 2 and 6 DF, p-value: 0.0001214

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Call:
lm(formula = SB_pct ~ Year + rule_change, data = yearly)

Residuals:
    Min       1Q   Median       3Q      Max
-1.8586 -0.6482  0.1133  0.6343  1.7485

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1107.9234   390.2911  -2.839   0.0296 *
Year           0.5864     0.1928    3.041   0.0228 *
rule_changePre -2.4608     1.4228   -1.729   0.1344
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.232 on 6 degrees of freedom
Multiple R-squared:  0.9013,    Adjusted R-squared:  0.8685
F-statistic: 27.41 on 2 and 6 DF,  p-value: 0.0009604

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By running linear regression, we observe an R-squared value of .95 for attempts, and .90 for success rate, meaning that the model explains a high percentage of the variation in change of variables. The rule change coefficient for attempts is -48.7 ($p=.000149$), meaning that following the rule changes, there are on average 48.7 more attempts per year compared to before, a statistically significant value indicating that the rule changes likely caused this increase in stolen base attempts. However, this same stat is only -2.46 ($p = .134$) for success rates, which is not statistically significant. This can be explained by observing the year coefficient for stolen base success rates (.587 $p=.023$), which is a statistically significant value meaning that success rate was already increasing by $\sim .59$ points per year before the rule changes, indicating a preexisting upward trend. When observing both stats together, we can conclude that rule changes caused a large, statistically significant increase in stolen base attempts, while the success rates were already increasing steadily prior to the rule changes and thus aren't necessarily correlated with the changes. This shows teams changed their strategies to be more aggressive on the bases as a result of the rule changes, but the gains in stolen base efficiency were due more to strategic growth rather than the effect of the rule changes.

Batted Balls Baserunning Aggression

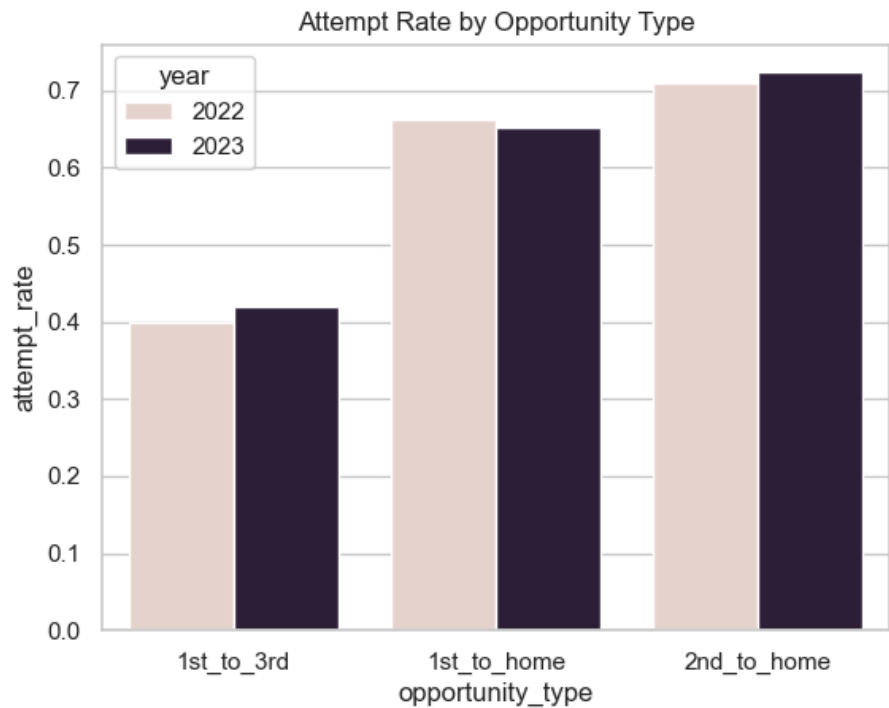
While stolen bases were a direct beneficiary of the 2023 rule changes, the implementation of the 18-inch bases and the shift restrictions created new incentives for baserunners. Below is a breakdown of extra-base attempt and success rates for 1st-to-3rd, 1st-to-home, and 2nd-to-home opportunities. To validate these trends, a Two-Sample Z-Test for Proportions was performed to determine if the changes between 2022 and 2023 were statistically significant ($p<0.05$).

Table 1: Raw data

year	opportunity_type	opportunities	attempts	successes	attempt_rate	success_rate
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2022	1st_to_3rd	6102	2441	2399	0.400033	0.982794
2022	1st_to_home	2567	1703	1658	0.66342	0.973576
2022	2nd_to_home	4832	3432	3338	0.710265	0.972611
2023	1st_to_3rd	5959	2506	2471	0.42054	0.986034
2023	1st_to_home	2560	1669	1628	0.651953	0.975434
2023	2nd_to_home	5025	3637	3564	0.723781	0.979929

Visualization of Base Running Attempts



Analysis of Attempt Rates:

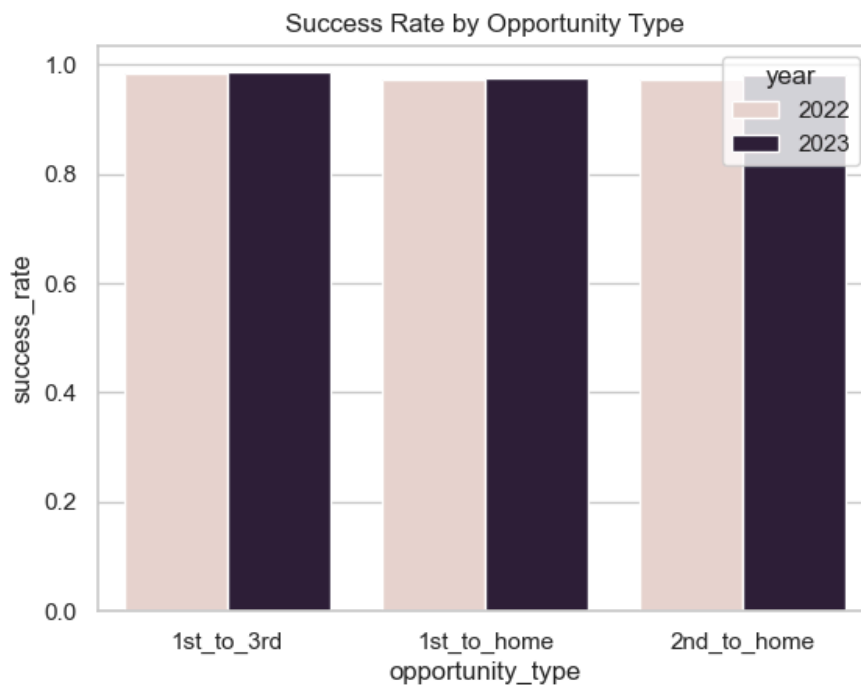
To evaluate if rule changes significantly altered the decision to "send" a runner, we compared the attempt rates using a Z-test. This allows us to distinguish genuine strategic shifts from random year-to-year variance.

Table 2: Attempts

Advancement type	Distance gained from base size increase	2022 Rate	2023 Rate
1st to 3rd	9 in.	40.0%	42.1%
2nd to Home	7.5 in.	71.0%	72.4%
1st to Home	12 in.	66.3%	65.2%

- **1st to 3rd:** This category saw the most significant growth at **2.1%**. This path covers two base-to-base segments, so it benefits from a cumulative 9-inch reduction in total distance (4.5 inches per segment). The p-value of **0.022** confirms this is a statistically significant strategic shift.
- **1st to Home:** Despite a 12-inch reduction in total distance, attempts slightly decreased by **1.1%**. Statistical testing (**p=0.387**) confirms this is not a significant shift in strategy. This scenario typically occurs on doubles where a 12-inch difference is often not a factor and the runner is usually clearly safe or clearly out.
- **2nd to Home:** Attempts increased by **1.4%**, reflecting a more aggressive approach when a runner is already in scoring position, though this did not reach the threshold for statistical significance (**p=0.136**).

Visualization of Base Running Successes



Analysis of Success Rates:

We also evaluated how this impacted baserunning materially by analyzing if runners were given an advantage outside of aggressiveness.

Table 3: Successes

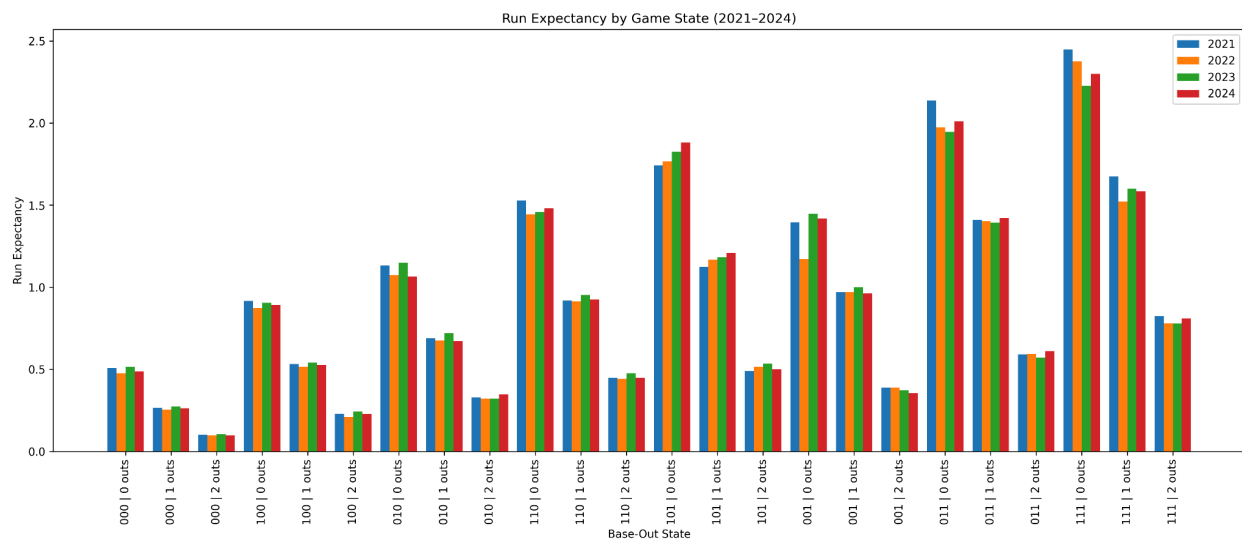
Advancement type	2022 Success Rate	2023 Success Rate	Difference
1st to 3rd	97.3%	98.0%	+0.73%
2nd to Home	98.3%	98.6%	+0.37%
1st to Home	97.4%	97.5%	+0.19%

We came to two primary conclusions based on this data. First, Success rates improved in every scenario during the 2023 season, despite the increase in attempt volume. Secondly, the uniform increase in success (notably the 0.73% jump for 1st-to-3rd) supports the hypothesis that the 18" bases provide a larger surface area, effectively reducing the likelihood of runners losing contact with the bag during high-speed slides.

The 2023 rule changes fostered a measurable increase in baserunning aggression, particularly on plays from 1st-to-3rd. Additionally, success rates uniformly increased significantly. The change in base size and shift restrictions definitively led to more base runner advancement. Further, teams successfully leveraged the changes to increase advancement frequency without sacrificing efficiency on the basepaths.

Run Expectancy

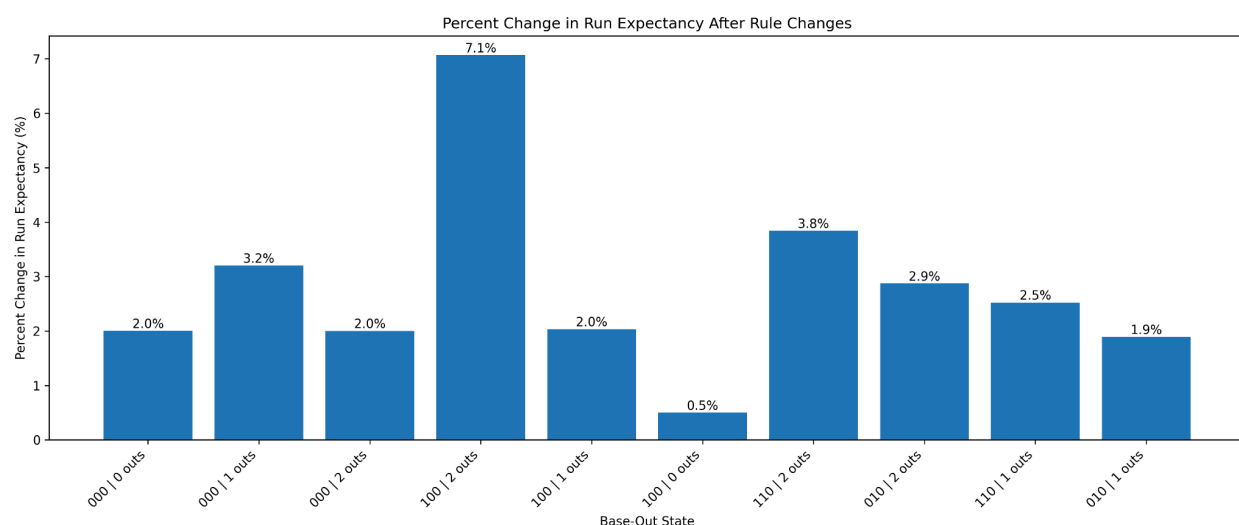
To further evaluate not just the changes in offensive behavior but the impact of the rule changes on offensive production, we analyzed run expectancy across all base-out states using data from the 2021-2024 seasons. Run expectancy measures the number of runs a team is expected to score from a given game state in the inning. For example, we can calculate the expected number of runs scored if there are runners on first and second with one out. This framework allows us to quantify how changes in baserunning behavior translate into overall scoring outcomes.



The bar plot above illustrates the run expectancy for each of the 24 game states, separated by year. As expected, states with more baserunners and fewer outs, such as bases loaded with no outs, have the highest run expectancy, while the state with no runners and two outs has the lowest.

Across years, the overall shape of the distribution remains stable, indicating that the fundamental structure of run scoring in baseball has not changed dramatically. However, there is

noticeable variation in specific states across seasons. This variation is most pronounced in relatively rare states, such as bases loaded with no outs or a runner on third with no outs. It also occurs between years with no rule changes. These fluctuations are likely driven by the smaller sample sizes. In contrast, more common states exhibit greater stability. To get a better sense of how the 2023 rule changes specifically impact run expectancy, the rest of our analysis will look at the 10 most commonly occurring game states, including traditional “steal situations.”

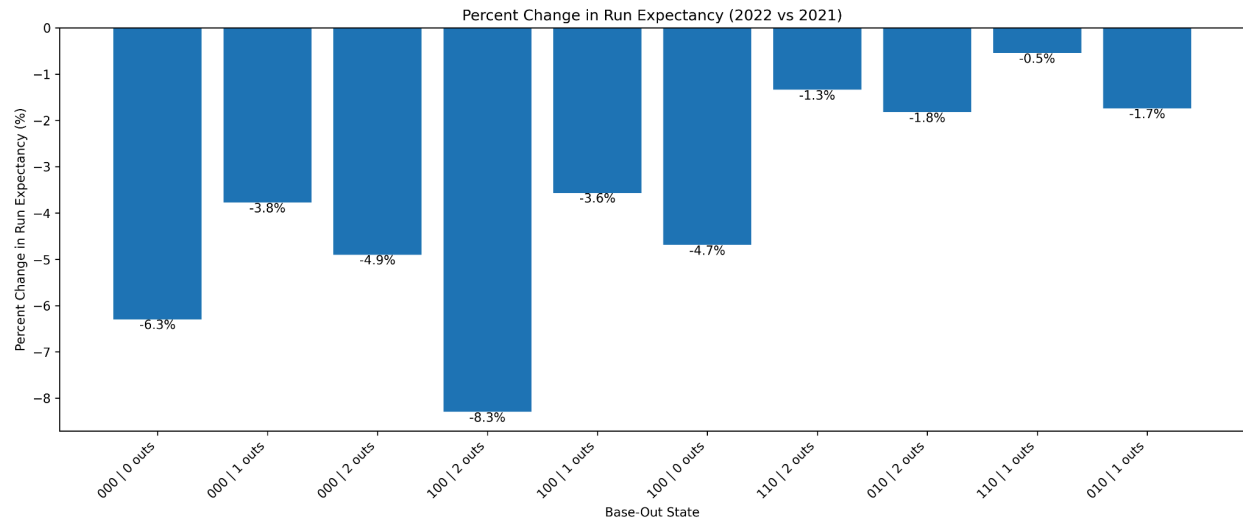


This plot displays the percent change in run expectancy for the most common game states, comparing pre-rule change data (2021 and 2022) to post-rule change data (2023 and 2024).

Within this subset, all states show modest positive changes in run expectancy following the rule changes. In particular, the state with a runner on first base and two outs exhibits a notable increase. This is the most common state to attempt a steal, and an increase in run expectancy is consistent with the expectation that rule changes designed to encourage baserunning would increase the likelihood of advancement and slightly improve scoring potential.

However, while these increases are all positive, they remain relatively small in magnitude. It is also important to note that run expectancy with the bases empty also increased, and in the case of 1 or 2 outs, it increased more than when a runner is on first. This suggests an increase in offensive production independent of baserunning. Other rule changes, such as banning the shift, could account for this.

To contextualize these findings, we examine percent changes in run expectancy between the 2021 and 2022 seasons, a period without rule changes. This serves as a baseline measure of natural fluctuations year to year.



Interestingly, all of the states had a negative change in run expectancy year-over-year, but the results show that changes between 2021 and 2022 are of similar magnitude to those observed between pre- and post-rule periods. This indicates that run expectancy is inherently variable across seasons. If we combine the percent changes, we find that the percent change between 2021 and the post-rule change era is flatter than the pre/post data.

To formally evaluate whether observed changes are statistically meaningful, we conduct hypothesis testing using a two-sample t-test comparing the distribution of runs remaining in each base-out state before and after the rule changes. The null hypothesis assumes no difference in run expectancy between periods.

In addition to p-values, we incorporate a calibration measure by comparing the magnitude of post-rule changes to baseline variation between 2021 and 2022. This “effect-to-noise” ratio provides a catch to test if observed changes exceed normal fluctuations, even if the change is statistically significant.

State	Outs	RE_Pre	RE_Post	Diff	Baseline Var (21→22)	Effect/Noise	p-value	Stat Sig (p<0.05)	Meaningful Effect
000	0	0.492	0.502	0.01	0.032	0.312	0.041	TRUE	FALSE
000	1	0.26	0.268	0.008	0.01	0.8	0.0401	TRUE	FALSE
000	2	0.099	0.101	0.002	0.005	0.4	0.4555	FALSE	FALSE
010	1	0.683	0.696	0.013	0.012	1.083	0.3955	FALSE	FALSE
010	2	0.326	0.335	0.009	0.006	1.5	0.3548	FALSE	FALSE
100	0	0.895	0.899	0.004	0.043	0.093	0.7404	FALSE	FALSE
100	1	0.524	0.534	0.011	0.019	0.579	0.2426	FALSE	FALSE
100	2	0.219	0.235	0.016	0.019	0.842	0.0086	TRUE	FALSE
110	1	0.916	0.939	0.023	0.005	4.6	0.2515	FALSE	FALSE
110	2	0.445	0.462	0.017	0.006	2.833	0.1964	FALSE	FALSE

If we look at states where a steal is possible, the results indicate that only one of them exhibits a statistically significant difference (runner on first with two outs). However, the change does not exceed the baseline variation. Thus, we conclude that the rule changes did not have a

meaningful effect in this regard. In fact, none of the states satisfy both statistical significance and the effect/noise ratio.

Overall, the analysis provides limited evidence that the 2023 rule changes produced a meaningful increase in run expectancy across base-out states. While certain situations show modest increases, these changes are small in magnitude and comparable to normal year-to-year variation without rule changes.

The baseline comparison between 2021 and 2022 demonstrates that run expectancy naturally fluctuates across seasons, sometimes even producing statistically significant differences in the absence of any structural changes to the game. When accounting for this variation, the observed post-rule changes appear less substantial.

These findings suggest that while the rule changes influenced player behavior and team strategy, particularly with respect to baserunning and stolen base attempts, they do not necessarily translate into an increase in offensive production. In other words, more aggressive baserunning strategies do not inherently translate to more runs.

This does not mean that the rule changes weren't impactful, as a primary motivation of the rules was to make the game more exciting. An increase in steal attempts and success rates achieves this objective. However, the impact on offensive production, and therefore the impact on winning, has not significantly changed.